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Towards First Detection of the Solar MSW Transition With JUNO

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Matter-induced neutrino flavor mixing (the Mikheyev-Smirnov-Wolfenstein, or MSW, effect) is a central prediction of the neutrino mixing framework, but it has not been conclusively observed. Direct observation of the energy-dependent MSW transition in the solar electron-neutrino survival probability would solve this, but backgrounds have been prohibitive. We show that our new technique for suppressing muon-induced spallation backgrounds will allow JUNO to measure the MSW transition at $>4\sigma$ significance in 10 years. This would strongly support upcoming multi-$1B next-generation long-baseline experiments and their goals in cementing the neutrino mixing framework.

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Introduction.— Neutrino flavor mixing in vacuum has been definitively established, showing that neutrinos have nonzero masses [1–12]. When neutrinos propagate through matter, their mixing can be substantially modified by the Mikheyev–Smirnov–Wolfenstein (MSW) effect [13, 14], which arises from coherent forward scattering via the weak interactions. Because these matter mixing effects depend on neutrino interactions at zero momentum transfer, they are highly sensitive to physics beyond the Standard Model (BSM) [15–28]. A quantitative experimental validation of matter mixing effects therefore remains an important goal.

However, no experiment has yet provided an unambiguous confirmation of matter mixing effects. For long-baseline and atmospheric experiments [29–33], it is challenging in part because Earth's column density is small. For solar experiments, the large and varying density of the Sun offers a powerful laboratory [34, 35]. *But while data on the electron-neutrino survival probability imply an MSW transition at a few MeV, it has never been directly observed* [6, 10, 17, 19, 36]. Further, measurements of the solar day-night effect, which also probe the MSW theory, are similarly inconclusive [6, 36].

A precise understanding of matter mixing effects is crucial not only for fundamental physics, but also for practical reasons. First, the Deep Underground Neutrino Observatory (DUNE) long-baseline experiment seeks to measure the neutrino mass ordering and the CP-violating phase [37, 38]; these measurements either depend on matter effects or are vulnerable to degeneracies with BSM scenarios [39–42]. (The Tokai to Hyper-Kamiokande, or T2HK, long-baseline experiment will also seek to measure the CP-violating phase, but under minimal matter effects [43].) Second, matter effects will be important for high-statistics measurements of atmospheric neutrinos in DUNE, T2HK, and other experiments [27, 31, 44, 45]. Third, matter effects are essential for interpreting observations of supernova neutrinos [46–48].

In this *Letter*, we show that the Jiangmen Underground Neutrino Observatory (JUNO) [49–51] — designed primarily to measure reactor neutrinos — can uniquely resolve this critical issue. In a companion paper [52], we present a powerful new background rejection technique for JUNO measurements of *solar neutrinos*. Exploiting this will give JUNO world-leading sensitivity to the MSW transition and will solidify the neutrino mixing framework. Figure 1 contrasts the current experimental status versus our projected sensitivity for JUNO.

Review of the MSW effect.— Neutrino mixing in vacuum can be described by a Hamiltonian evolution equation [53, 54]. This depends upon the mixing angles between the flavor and mass eigenstate bases (θ_{ij}) and the mass-squared differences for the mass eigenstates (Δm^2_{ji}) [55]. When neutrinos propagate in matter, the Hamiltonian is modified by adding the MSW term ("the matter potential") for electron neutrinos. This term is $\propto G_F N_e$, where G_F is the Fermi constant and N_e is the electron density in the medium [13, 14, 19, 27, 53, 54]. For solar neutrinos of energy E_ν, the electron-neutrino survival probability is [19, 56]

$$P_{ee}(E_\nu) \simeq \sin^4(\theta_{13}) + \cos^4(\theta_{13})\, P^{2\nu}_{ee}(E_\nu), \qquad (1)$$

$$P^{2\nu}_{ee}(E_\nu) = \frac{1}{2}[1 + \cos(2\theta_{12})\langle\cos(2\theta_m)\rangle], \qquad (2)$$

where $\cos(2\theta_m)$ refers to the matter mixing angle, and this term is averaged over the solar production region [57, 58]. The energy dependence of Eq. (2) — which provides a direct probe of matter mixing effects — arises from θ_m, which depends on the vacuum mixing angle and a dimensionless quantity $\propto G_F N_e E_\nu/\Delta m^2_{21}$. The angle θ_m is θ_{12} in vacuum (or low neutrino energies), $\pi/4$ at the MSW resonance density (or energy), and $\pi/2$ at high density (or high neutrino energies).

Figure 1 shows the expected survival probability in the standard MSW framework using mixing parameters from the latest global fit [59] and recent JUNO measurements [51]. At the low energies probed by Borexino measurements (using *pp*, *pep*, and ^{7}Be neutrinos) [10], the results are consistent with the vacuum limit of Eq. (2).

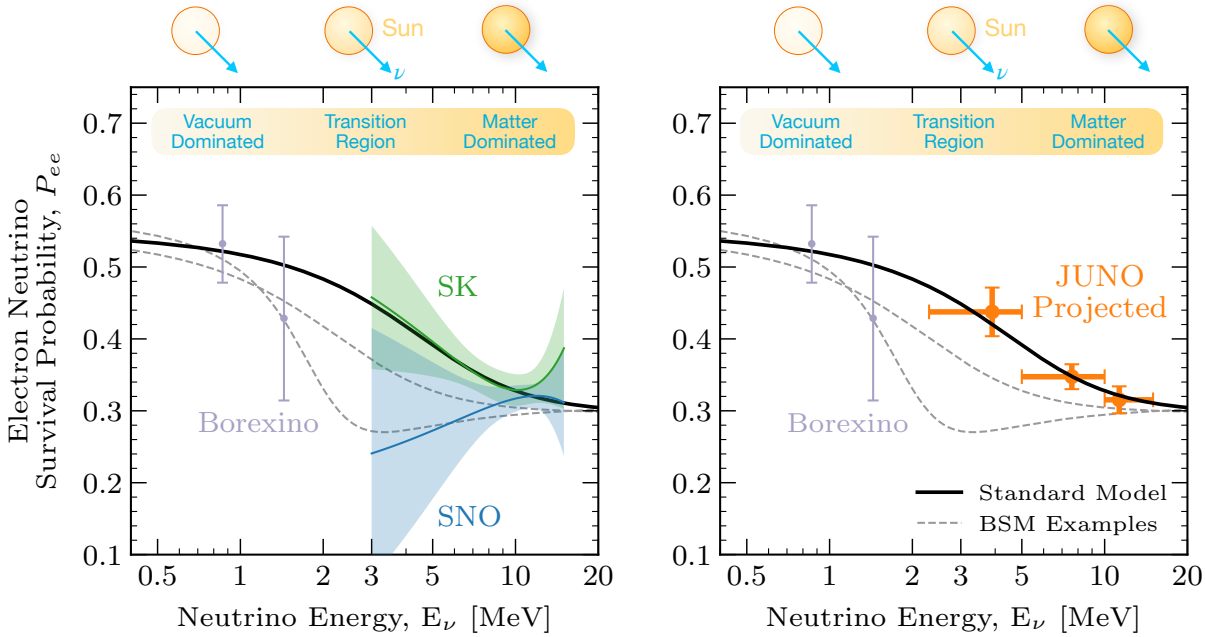

FIG. 1. Comparison of current [6, 10, 36] and projected measurements (using our background-rejection techniques for JUNO) of the solar MSW transition. The black line shows the MSW prediction, while the gray dashed lines show example allowed NSI models. The bins shown for JUNO are a projection, assuming the Standard Model, of how they could go beyond the quadratic fits used by SK and SNO. *With our spallation cuts, JUNO can measure the MSW transition well.*

At the high energies probed by Super-Kamiokande (SK) and Sudbury Neutrino Observatory (SNO), the results are consistent with the high-density limit [6, 36]. In the few-MeV transition region, limited statistics lead to large uncertainties: SK data weakly favor an *upturn* (at 1.2σ), while SNO data weakly favor a *downturn* (at 0.7σ). When taken together they somewhat favor an *upturn* (at 2.1σ) [4, 36]. These measurements are thus not yet sufficient to test matter mixing effects.

Figure 1 also shows that the survival probability can be modified by BSM physics, where we focus on non-standard interactions (NSI) induced by new heavy mediators [17, 19, 20, 22, 25–28, 60]. Other possibilities include neutrino decay, sterile neutrinos, and large neutrino magnetic moments [15, 16, 18, 19, 21, 24]. For NSI, the matter potential is modified by new interactions between neutrinos and fermions, adding a new term that is $\propto G_F N_f \epsilon^f_{\alpha\beta}$, where N_f is the density of the fermion f in the medium and $\epsilon^f_{\alpha\beta}$ are dimensionless parameters that quantify the strength of the NSI between that fermion and the neutrino flavors ν_α and ν_β [20, 27]. We consider two example NSI models that are allowed at the 3σ level, corresponding to the parameters $(\epsilon_D, \epsilon_N) = (-0.20, -0.30)$ (bottom line) and $(-0.15, -0.12)$ (top line) as defined in Refs. [27, 61]. These are equivalent to $\epsilon^d_{ee} \sim 0.4$ and $\epsilon^d_{e\tau} \sim 0.1$.

Prospects for solar neutrinos in JUNO.— Of present and planned detectors, JUNO has the best potential to make precise measurements of the MSW transition. (For comparison, Hyper-Kamiokande expects to have a sensitivity of 3σ in 10 years with an expected threshold of 4.5 MeV but may do better [43, 45].) JUNO detects solar neutrinos through scattering with electrons, where the recoiling electron produces a prompt scintillation signal. The large scintillation light yield provides excellent energy resolution ($3\%/\sqrt{(E/\mathrm{MeV})}$) and enables a low detection threshold [50, 51, 62], both essential for detecting the MSW transition. With a 20-kton target mass, nearly 70 times larger than Borexino, JUNO will detect ^{8}B neutrinos at a rate of ∼80 events per day [50, 63].

JUNO's analysis threshold for ^{8}B solar neutrinos is set by detector backgrounds. One important source is the natural radioactivity originating from contaminants in the scintillator or surrounding material. These backgrounds are effectively mitigated through multiple purification techniques and fiducial volume cuts [50]. Initial performance results [62] indicate that the concentrations of uranium and thorium in the detector are within the levels necessary to achieve a 2.3-MeV threshold, the lowest to date. A somewhat high activity rate of polonium may be a challenge, but it should fall to the required level within roughly one year, based on its 138.4-day half-life.

For the analysis window of 2.3–15 MeV, the dominant background is caused by the delayed decays of radioactive isotopes produced by cosmic-ray muons through spallation (nuclear breakup) processes [50, 52]. The rate of these decays is ∼4000 events per day. The standard approach to reducing spallation backgrounds is to veto all events within a cylindrical region following a cosmic-ray muon. In Ref. [50], JUNO defined baseline cuts that would reduce spallation backgrounds to the sub-percent level, but at the cost of 50% signal loss.

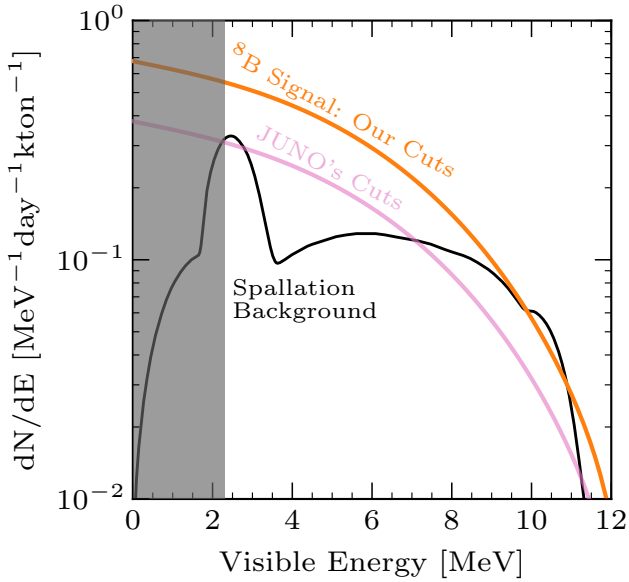

FIG. 2. The ^{8}B signal spectrum at JUNO using our neutron-tagged cuts compared to using JUNO's cuts [50], both compared to the remaining spallation background spectrum. *With our spallation cuts, twice more signal would be retained.*

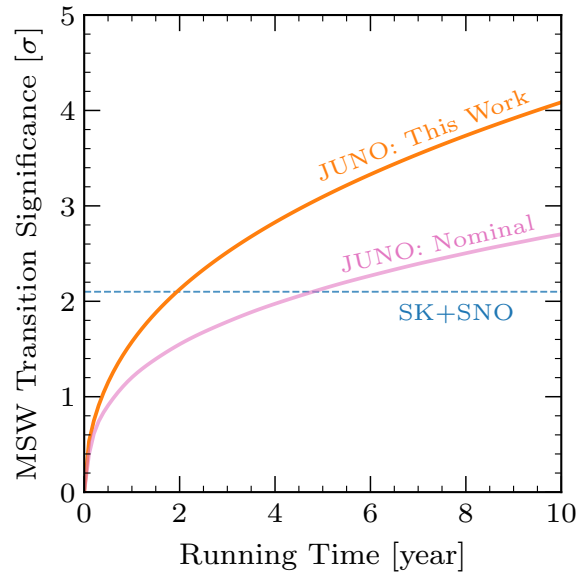

FIG. 3. Comparison of JUNO's sensitivity to the solar MSW transition with our cuts and without, as well as to the combined SK+SNO result [36]. *With our spallation cuts, greater significance would be reached, and more quickly.*

New technique to reduce backgrounds.— For JUNO to achieve its full potential as a solar neutrino detector, improved background rejection techniques are required. In earlier papers on SK [64–67], we showed that nearly all spallation isotopes are produced in muon-induced hadronic showers through the secondary particles associated with them, such as charged pions and neutrons. Importantly, hadronic showers are rare and are produced by $\lesssim$5% of muons. These considerations are general to other detectors [52, 68]. This motivates a strategy with aggressive cuts on hadronic showers and moderate cuts on electromagnetic showers.

In a companion paper [52], we demonstrate that secondary neutrons are an efficient signature to tag hadronic showers in JUNO. These neutrons scatter down to thermal energies and capture on hydrogen, emitting a 2.2 MeV gamma ray, which JUNO detects with nearly perfect efficiency. For hadronic showers, we find an average yield of ∼14 neutrons, and at least one neutron 98% of the time. For hadronic showers, we apply cylindrical cuts with radii up to 6 m and times up to 40 s. For electromagnetic showers, we instead limit these to 3 m and 5 s. This strongly suppresses spallation backgrounds while keeping the deadtime (signal loss) minimal.

Figure 2 shows the increase in the ^{8}B signal rate at JUNO using our neutron-tagged cuts (for the details of this figure, see Ref. [52]). For the same background reduction efficiency, our method reduces the deadtime by a factor of ∼5 compared to JUNO's baseline approach (from ∼50% to ∼10%), yielding a factor of ∼2 gain in exposure. This improvement in the signal-to-background ratio is critical for testing the MSW transition.

Probing the MSW transition.— To test sensitivity to matter mixing effects, one must compare simulated data in JUNO under two hypotheses:

1. P_{ee} = constant (but allowed to vary).
2. P_{ee} given by the MSW theory in Eq. (2).

With JUNO's baseline cuts, they could reject the constant-probability hypothesis at 2.7σ significance in 10 years [50], somewhat better than the present combined SK+SNO result of 2.1σ.

We calculate the recoil electron spectrum for each case by convolving the ^{8}B neutrino spectrum [69] with the elastic neutrino-electron cross section, including radiative corrections [70], and the appropriate survival probability. There is a minor gain from the fact that the ^{8}B spectrum from Ref. [69] has reduced shape uncertainties compared to the one adopted in Ref. [50]. We incorporate detector effects, including the energy resolution (∼3% at 1 MeV) energy-scale uncertainty (∼1%) [50, 62], detection efficiency [50], and fiducial volume cuts [50]. We simulate the detector backgrounds, including spallation isotopes, intrinsic radioactivities, and reactor neutrino elastic scattering [50]. We take into account statistical uncertainties as well as appreciable systematic uncertainties on the flux [6], spectrum shape [69], and backgrounds [50].

To quantify JUNO's discrimination power, we follow the approach of their Ref. [50]. Assuming the MSW case as the truth and using bins of width 0.1 MeV, we evaluate the hypothesis of the flat-probability case using a χ^2 test, taking into account all uncertainties. Our analysis is slightly more conservative than JUNO's, as we start at 2.3 MeV instead of 2.0 MeV to take into account a new spallation background we identified (see Ref. [52]).

Figure 3 shows that JUNO, with improved spallation cuts and assuming the Standard Model, can reject the flat survival probability hypothesis with $>4\sigma$ significance in 10 years. Moreover, JUNO will surpass the current SK+SNO result in about two years, less than half the time needed with JUNO's baseline cuts. Of course, JUNO results may also favor BSM scenarios. This highlights the significance of our new background rejection techniques [52]. We expect that our results can be improved further by developing likelihood-based cuts and employing machine learning techniques [71, 72].

As an aside, in Fig. 1, we illustrate how JUNO could measure the survival probability in mostly uncorrelated bins, with different systematics dominating the uncertainties in each. This approach is in contrast to the quadratic fit bands used by SK and SNO. In calculating these bins, we follow Refs. [6, 36] in extracting the electron-neutrino survival probability by taking into account the contribution of muon and tau neutrinos.

Implications for future experiments.— Measurements of long-baseline neutrino flavor mixing aim to determine, among other goals, the neutrino mass ordering and the value of the CP-violating phase. At DUNE, the ability to perform these measurements is linked directly to the presence of matter effects. If BSM physics alters neutrino propagation in matter, it could pose significant challenges for these measurements. For example, NSI can substantially degrade the mass-ordering sensitivity [42, 73–75] or even lead to errant interpretations of CP violation in long-baseline neutrino mixing [41, 76–79].

To quantify the impact of matter-effect degeneracies on DUNE's mass-ordering reach, we simulate expected event rates (in electron- and muon-neutrino samples) assuming six years of data collection — equal running in neutrino and antineutrino modes — with a detector fiducial mass of 20 kton. Mixing parameters are fixed to the best-fit values from Ref. [59] (with solar parameters updated from Ref. [51]), with a varying assumed-true $\delta_{\rm CP}$. For the DUNE Baseline scenario, we allow θ_{23}, $\delta_{\rm CP}$, and Δm^2_{31} (fixing $\Delta m^2_{31} < 0$) to vary, obtaining a best-fit χ^2.

We consider representative NSI scenarios where the additional effective electron neutrino coupling is $\epsilon_{ee} = \sum_f \epsilon^f_{ee} N_f / N_e$. Current global constraints on ϵ_{ee} are model-dependent and vary depending on the mediator mass, but allow values roughly in the range $(-1, 1)$ at the 3σ level [26, 27]. A precise measurement of the expected MSW transition in JUNO would restrict ϵ_{ee} to the range $(-0.2, 0.2)$ at 1σ and $(-0.3, 0.4)$ at 3σ.

Figure 4 illustrates how JUNO's measurement of the MSW transition can mitigate degeneracies that affect DUNE's sensitivity to the mass ordering. When all $\epsilon = 0$ (the Standard Model), DUNE will reach a significance of 7–15σ, depending on the value of the CP-violating phase, $\delta_{\rm CP}$. Allowing NSI within currently allowed bounds introduces degeneracies that reduce this sensitivity to being below 5σ for the majority of $\delta_{\rm CP}$ values, which would undermine DUNE's goals. However, by independently probing matter mixing effects through solar neutrinos, JUNO will lift these degeneracies and restore DUNE's mass-ordering sensitivity to more than 5σ for all $\delta_{\rm CP}$.

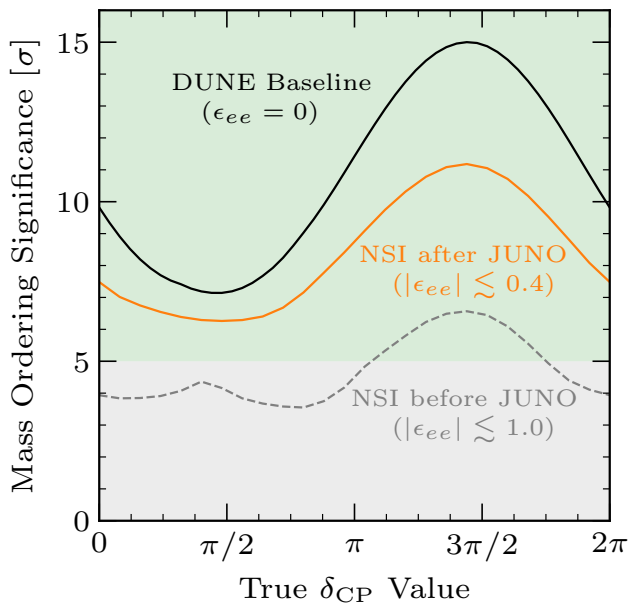

FIG. 4. DUNE sensitivity to the neutrino mass ordering under different assumptions: Standard Model only, NSI within a currently allowed range, and NSI constrained by JUNO's measurement of the MSW transition. *JUNO's measurement of matter mixing effects will support DUNE's success.*

Conclusions.— We have shown that JUNO can provide a decisive experimental test of the MSW transition in solar neutrino mixing. By exploiting neutron tagging to muon-induced hadronic showers, spallation backgrounds in the critical 2.3–15 MeV energy range can be substantially reduced, enabling a measurement of the MSW transition with a significance $>4\sigma$ in 10 years.

Long-baseline accelerator experiments such as DUNE provide a concrete example of the importance of measuring the solar MSW transition, where JUNO alone can break degeneracies that limit their interpretation. In turn, this would provide a firm experimental foundation for next-generation measurements of the neutrino mass ordering and CP violation. The JUNO measurement could be strengthened by combining with other neutrino mixing measurements in a global analysis.

Beyond this, measuring the MSW transition would provide a model-independent calibration of matter effects, placing robust constraints on new physics that modifies neutrino propagation in matter. By doing so, a JUNO observation of the MSW transition would directly inform the interpretation of neutrino mixing across accelerator, atmospheric, and supernova experiments.

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